Why I think this is an excellent MATH3888 baseline

Your starting objective could be something like:

> **Reproduce the experimentally observed TTX-induced slow-wave oscillations of R15 across different applied-current conditions.**

Then mathematically study why the transition occurs.

For example:

$$I_{\text{app}} \quad \longrightarrow \quad \begin{cases} \text{stable equilibrium} \\ \text{stable periodic orbit} \\ \text{stable equilibrium} \end{cases}$$

The 1975 paper itself reports precisely this sort of behaviour in its exemplary model: stable equilibria for some parameter values and instability/limit-cycle behaviour around an intermediate value.

Bursting of R15 Cell

That gives you an extremely natural continuation parameter.

You could then ask:

$$\text{At what parameter values do the equilibria lose/regain stability?}$$

and use PyDSTool to identify the associated bifurcations.

That's much closer to the type of project you've been practising for than I initially feared.

Mathieu & Roberge 1971 on PubMed ↗

Agent / manipulation	Paper	What it does experimentally	Role for our project
TTX (tetrodotoxin)	Plant & Kim 1975/76; Junge & Stephens 1973	Blocks the fast spike-generating inward/Na current, so action potentials disappear and the underlying slow wave becomes visible	Main baseline perturbation
Ouabain	Junge & Stephens 1973	Inhibits the Na/K electrogenic pump	Tests whether the pump *causes* the rhythmic hyperpolarization
DNP, lithium, K-free saline, cooling	Junge & Stephens 1973	Other ways of interfering with metabolic/electrogenic-pump activity	Alternative tests of the pump hypothesis
TEA	Junge & Stephens 1973	Blocks some K currents	Helps identify what kind of K conductance is responsible
Serotonin / 5-HT	Levitan & Levitan 1988	Neuromodulator acting through cAMP; changes K and Ca currents and therefore bursting	Excellent possible project extension
Dopamine	Levitan & Levitan 1988	Modulates R15 and attenuates part of the 5-HT response	Possible *two-drug* extension
Forskolin / 8-Br-cAMP / IBMX etc.	Levitan & Levitan 1988	Manipulate the cAMP pathway	Mechanistic controls rather than something I'd model initially

My current preference before your pitch

Given this dimensionality constraint, I'd **not replace Plant with Canavier outright**.

I'd think of them as:

$$\boxed{\text{Plant} = \text{manageable mathematical base}}$$

and

$$\boxed{\text{Canavier} = \text{richer physiological reference / source of extensions.}}$$

Then the group can decide whether your final base model is:

- Plant's 5D TTX model;
- Plant's 4D 1981 model;
- or a deliberately reduced 3–5D model extracted from Canavier.

That distinction probably needs to be settled **before** you promise a model in the pitch, because you're completely right: the phrase “11-state model” sounds very different from the 3–5D ODE project the professor described.